Fiscal Year:	FY 2026	**Task Last Updated:**	FY 03/27/2026
PI Name:	Moudy, Sarah Ph.D.		
Project Title:	Development of Sensorimotor Fitness for Duty Assessments Using Ground Analogs		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/27/2022	**End Date:**	09/30/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	This investigation is a continuation of a Directed Research project with the same name under PI Marissa Rosenberg. Dr. Rosenberg has left NASA. The new PI is Sarah C. Moudy, Ph.D. with KBR/Aegis Aerospace. Millard Reschke, Ph.D. has retired from NASA so he no longer serves as Co-Investigator.		
COI Name (Institution):	Wood, Scott Ph.D. (NASA Johnson Space Center) Peters, Brian Ph.D. (NASA Johnson Space Center) Clark, Torin Ph.D. (University of Colorado, Boulder) Schubert, Michael (Johns Hopkins University)		
Grant/Contract No.:	Directed Research		
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Task Description:	The first aim developed a Sensorimotor Disorientation Analog (SDA) that provided different levels of acute disorientation through combined vestibular, visual, and proprioceptive disruptions. The SDA was used to increase the range of performance, simulating the moderate-to-severe performance decrements observed postflight. The levels of the SDA were titrated and validated by (1) using feedback from experienced astronauts and (2) comparison to tasks that have a wealth of spaceflight data at different time points during recovery. Astronaut postflight experience provided a more realistic starting point for validation of the SDA. Referencing existing postflight data also helped characterize how each magnitude of the SDA compared to recovery from long-term microgravity exposure. The second aim assessed the suitability of a proposed set of sensorimotor assessment tasks, or measures that were feasible within the limited time, resources, and space of a lunar/Martian lander. The assessment tasks aided in progressive adaptation to the novel gravitational environment, provided opportunities to develop strategies to recover from off-nominal body positions, and mimicked operational tasks such that crew could self-assess their potential ability to complete their missions. We obtained performance of the sensorimotor assessment tasks at varying levels of SDA magnitude to map sensorimotor ability to the probability of completion of operational analog tasks. The third aim involved 60 minutes of prolonged +3Gx centrifugation to mimic the vestibular alterations observed after gravity transitions. Immediately after egress from the centrifuge, subjects performed the sensorimotor assessment tasks followed by the operational analog tasks. Performance was mapped between the assessment and operational tasks, similar to Aim 2. The fourth aim will characterize the effects of reduced gravity load on the sensorimotor assessment tasks to define performance thresholds that would indicate crew readiness for operational tasks. This will be defined using a parabolic flight. The fifth and final aim will build on the lunar landing simulations that have be developed for the Manual Piloting flight study (PI: Wood) and the human lander systems training simulations to support the Flight Operations Directorate for exploration class mission training. The study will involve familiarization sessions focused on developing proficiency to perform the landing task. Following a standard duration mission timeline (~6 months), the same lunar landing will be performed while experiencing sensorimotor disorientation (galvanic vestibular stimulation). Between the familiarization and final test sessions, subjects will undergo 1 of 3 “inflight” training paradigms using an onboard trainer including: 1) no training, 2) consistent training throughout the “mission”, or 3) training only at the end of “mission” consistent with current International Space Station (ISS) protocols. The outcome of this aim is to characterize the relationship between performance on the onboard training platform with disoriented lunar landing performance to inform what performance threshold is required to mitigate risk associated with vertigo and disorientation with manual crew override during landings. The main deliverables from this project will be recommended sensorimotor assessments to be completed prior to extravehicular activity (EVA) operations that provide a quantitative index of readiness to perform key operational tasks and validation of lunar landing simulations for training manual crew override during landings.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This project is in direct response to the baselined Human Research Program (HRP) Path to Risk Reduction milestone of providing updates to the NASA Fitness For Duty Standards. The new standards should be tied to fitness for duty for exploration tasks and provide a quantitative index of readiness to perform key exploration tasks. This research effort will leverage expertise based upon HRP-funded flight research investigations including Functional Tasks Test, Field Test, Standard Measures, and Manual Control, as well as MedB computerized dynamic posturography. This project will leverage critical mission tasks previously established by Ryder et al. ("A novel approach for establishing fitness standards for occupational task performance." Eur J Appl Physiol, 2019) for standards related to the risk of reduced muscle mass, strength, and endurance. This project must also leverage experience with vestibular spaceflight analogs (e.g., Galvanic vestibular stimulation, sustained 3Gx centrifugation) to characterize how the deconditioned state following G-transitions (e.g., postural instability, motion sickness, head movement restrictions) map to functional performance.
Research Impact/Earth Benefits:	This research addresses the Risk of Altered Vestibular/Sensorimotor Function Impacting Critical Mission Tasks (Sensorimotor Risk) and associated research gaps from NASA’s Human Research Roadmap. The development of sensorimotor assessments for EVA and manual control addresses each gap associated with the characterization of the effects of spaceflight on the sensorimotor and neuro-vestibular systems, including postural control and locomotion (Gap SM-101), manual control (Gap SM-102), spatial orientation and motion sickness (Gap SM-103), and other brain functions such as sleep, cognition, attention (Gap SM-104). This project is in direct response to the baselined Human Research Program (HRP) Path to Risk Reduction milestone of defining sensorimotor assessments that can provide a quantitative index of readiness to perform key exploration tasks. The Flight Operations Directorate has identified the need for better sensorimotor assessment tools that would protect against hazards of performing operational tasks too soon following gravitational transitions. The project outcomes will address this recommendation by providing a validated set of sensorimotor assessment tasks that will 1) allow for self-assessment when communication with ground support may be limited or delayed and 2) provide progressive adaptation to the novel gravitational environment. For Earth-based benefits: The sensorimotor community has a need for simple assessment tasks that could be used clinically, through telemedicine, or by the patient alone. These tasks could be used to help in self-determination of the patient’s sensorimotor ability to perform daily activities and/or guide recovery of vestibular-impaired patients.
	During this reporting period, Aim 5 testing began in June 2025 and is expected to be completed by October 2026. Aim 5 Methods As mission duration increases, lack of training retention is expected to negatively impact operator performance of complex tasks (1, 2). This risk may be exacerbated during periods of G-transitions since vestibular alterations can impact time perception, attention, and cognitive processing ability (3), especially related to spatial memory (4). For longer duration Space Shuttle flights, a "Portable In-flight Landing Operations Trainer" (PILOT) laptop computer simulator was flown as a tool for helping the mission commander and pilot maintain their proficiency for approach and landing (5). One Shuttle commander, who experienced vertigo during the landing (personal communication), attributed the learned ability to lower perceived cognitive workload and effectively pilot through disorientation to complete a successful landing to the onboard training using PILOT. Therefore, the purpose of this study aim is to establish the onboard training protocols required to maintain piloting proficiency and minimize any impacts associated with sensorimotor disorientation for planetary surface landings. A total of 60 subjects will be allocated into one of three training groups (20 subjects per group): 1) consistent onboard training at a frequency of once per month, 2) onboard training only conducted within the month prior to end of “mission”, and 3) no onboard training. As of March 2026, all 60 subjects have begun testing, of which 26 subjects have completed testing, 26 subjects

Task Progress:	are active and assigned to a training group, and 8 subjects are undergoing proficiency training. A total of 286 test sessions have been completed to date. In preparation for testing, a new galvanic vestibular stimulation (GVS) profile was developed. GVS, as a disorientation paradigm, has demonstrated the ability to replicate Shuttle and ISS astronaut postflight functional performance in tasks targeting postural and dynamic stability (6, 7). The GVS profile developed for Aims 1-3 targeted functional gross motor performance. However, a light finger touch on a stationary surface is able to provide enough postural stability to complete a functional task (e.g., hatch opening) postflight similarly to preflight (8, 9). Lunar landing pilots will have structural support (e.g., arm rests) that likely will minimize gross motor performance decrements. Therefore, a GVS profile that targets manual control performance was needed, focusing on general vestibular disruption, alterations in visual acuity, and increases in cognitive demand. GVS has been used in landing simulators and shown to replicate pilot performance during Shuttle landings, demonstrating the impact of GVS on manual control (10). The frequencies used in the Moore et al. (2011) study GVS profile provided the base with which comparisons could be made. As piloting tasks involve slow movements, a new GVS tester profile was created to include lower frequencies, especially those below 0.43Hz, as decreases in postflight motion perception have been found in ISS astronauts (11). Pilot testing was conducted on 5 subjects to compare performance with and without GVS on the nulling task. The order of GVS profiles was counterbalanced. The results showed that GVS did minimally impact performance, compared to without GVS, and that consistently, the new profile with the additional low frequency components has slightly more impact than the Moore et al. (2011) profile. Anecdotal feedback suggested that GVS increased the cognitive demand, even if manual control performance was not greatly impacted. These results provided evidence to include the new profile in Aim 5 testing. References 1. Foale CM, Kaleri AY, Sargsyan AE, Hamilton DR, Melton S, Martin D, Dulchavsky SA. Diagnostic instrumentation aboard ISS: just-in-time training for non-physician crewmembers. Aviation, space, and environmental medicine. 2005;76(6):594-8. 2. Dempsey DL, Barshi I. Applying research-based training principles: Toward crew-centered, mission-oriented space flight training. Psychology and human performance in space programs: CRC Press; 2020. p. 63-80. 3. Bigelow RT, Agrawal Y. Vestibular involvement in cognition: Visuospatial ability, attention, executive function, and memory. Journal of vestibular research. 2015;25(2):73-89. 4. Smith PF. Hearing loss versus vestibular loss as contributors to cognitive dysfunction. Journal of neurology. 2022;269(1):87-99. 5. Kennedy RS, Lane NE, Turnage JJ, Harm DL, editors. Individual Differences in Visual, Motor and Cognitive Performances: Correlations with a Simulated Shuttle Landing Task. Proceedings of the Human Factors and Ergonomics Society Annual Meeting; 1997: SAGE Publications Sage CA: Los Angeles, CA. 6. Moudy SC, Peters BT, Clark TK, Schubert MC, Wood SJ. Development of a ground-based sensorimotor disorientation analog to replicate astronaut postflight experience. Frontiers in Physiology. 2024;15:1369788. 7. Moore ST, MacDougall HG, Peters BT, Bloomberg JJ, Curthoys IS, Cohen HS. Modeling locomotor dysfunction following spaceflight with Galvanic vestibular stimulation. Experimental brain research Experimentelle Hirnforschung Experimentation cerebrale. 2006;174(4):647-59. Epub 2006/06/10. doi: 10.1007/s00221-006-0528-1. PubMed PMID: 16763834. 8. Mulavara AP, Peters BT, Miller CA, Kofman IS, Reschke MF, Taylor LC, Lawrence EL, Wood SJ, Laurie SS, Lee SMC, Buxton RE, May-Phillips TR, Stenger MB, Ploutz-Snyder LL, Ryder JW, Feiveson AH, Bloomberg JJ. Physiological and Functional Alterations after Spaceflight and Bed Rest. Med Sci Sports Exerc. 2018;50(9):1961-80. Epub 2018/04/06. doi: 10.1249/MSS.0000000000001615. PubMed PMID: 29620686; PMCID: PMC6133205. 9. Lackner JR, DiZio P, Jeka J, Horak F, Krebs D, Rabin E. Precision contact of the fingertip reduces postural sway of individuals with bilateral vestibular loss. Experimental brain research. 1999;126:459-66. 10. Moore ST, Dilda V, MacDougall HG. Galvanic vestibular stimulation as an analogue of spatial disorientation after spaceflight. Aviat Space Environ Med. 2011;82(5):535-42. Epub 2011/05/28. doi: 10.3357/asem.2942.2011. PubMed PMID: 21614868. 11. Moore ST, Dilda V, Morris TR, Yungher DA, MacDougall HG, Wood SJ. Long-duration spaceflight adversely affects post-landing operator proficiency. Sci Rep. 2019;9(1):2677. Epub 2019/02/26. doi: 10.1038/s41598-019-39058-9. PubMed PMID: 30804413; PMCID: PMC6389907.
Bibliography Type:	Description: (Last Updated: 04/01/2026)
Conference Materials (Downloadable)	Moudy SC, Peters BT, Clark TK, Schubert MC, Wood SJ. "Predicting spaceflight operational performance from sensorimotor assessment tasks." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026